

Understanding Transcriptional Regulatory Redundancy by Learnable Global Subset Perturbations

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Cis-Regulatory Elements and Transcriptional Regulatory

Cis-Regulatory Elements (CREs) are regions of DNA that regulates the expression of genes

- CREs is **crucial for numerous biological functions**, with its disruption potentially leading to various diseases
- CREs often **exhibit redundancy**, allowing them to compensate for each other in response to external disturbances
- Single-cell sequencing technologies offer researchers an opportunity to understand this regulatory redundancy

Notations

ATAC-seq $\mathbf{x} \in \{0, 1\}^{d_a}$ ($d_a > 120k$), where each dimension of this vector indicates the peak state in chromosomes

RNA-seq $\mathbf{y} \in \mathbb{R}^{d_r}$, gene expression values regulated by the ATAC-seq

Transcriptional Regulatory A complex gene regulatory process $\mathbf{y} = \mathcal{F}(\mathbf{x})$, where $\mathcal{F}$ is a **black-box** function.

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Transcriptional Regulatory Redundancy Problem

Regulatory Redundancy Problem Given a **target gene in y** , what **combinations of entries in x** are regulating the expression of the target gene.

Examples

Let's say we have 4 cells, and a global removal is applied.

$$\mathbf{X} = \begin{bmatrix} 1 & \cdots & 1 & 0 \\ 0 & \cdots & 0 & 1 \\ 1 & \cdots & 0 & 0 \\ 0 & \cdots & 1 & 1 \end{bmatrix} \xRightarrow{\mathcal{F}} \mathbf{y} = \begin{bmatrix} 0.7 \\ 0.1 \\ 2.2 \\ 0.9 \end{bmatrix} \xRightarrow{\text{After}} \tilde{\mathbf{X}} = \begin{bmatrix} 0 & \cdots & 0 & 0 \\ 0 & \cdots & 0 & 1 \\ 0 & \cdots & 0 & 0 \\ 0 & \cdots & 0 & 1 \end{bmatrix} \xRightarrow{\mathcal{F}} \tilde{\mathbf{y}} = \begin{bmatrix} 0.1 \downarrow \\ 0.0 \downarrow \\ 0.6 \downarrow \\ 0.2 \downarrow \end{bmatrix}$$

Challenges

- The complex gene regulatory process function $\mathcal{F}$ is an indifferentiable black box
- The combinatorial effect of multiple CREs is important but has been ignored by most previous methods
- $\mathbf{x}$ is **sparse, discrete, and high-dimension**. The combinatorial solution space of $\text{subset}(\mathbf{x})$ is vast, which means brute-force search is intractable
- **In this work, we propose GRIDS to solve the above challenges**

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We train a differentiable surrogate $\hat{\mathcal{F}}$ to mimic the black-box $\mathcal{F}$

- a neural network to act as a differentiable surrogate $\hat{\mathcal{F}}$
- $\hat{\mathcal{F}}$ can predict the transcriptional regulatory process (an emulator to the black-box regulatory function $\mathcal{F}$).

$$\mathbf{y} = \hat{\mathcal{F}}(\mathbf{x})$$

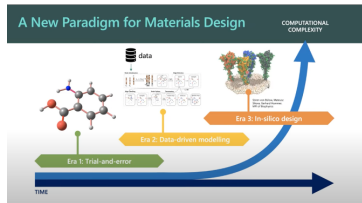


Figure: Deep Learning and the Natural Sciences: A Perfect Marriage?

Questions

- It requires an accurate surrogate model $\hat{\mathcal{F}}$ to $\mathcal{F}$
- Can we understand the transcriptional regulatory redundancy problem by interpreting the surrogate $\hat{\mathcal{F}}$?

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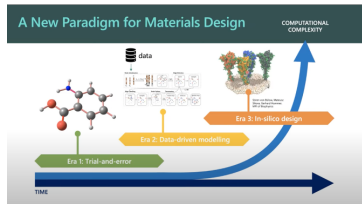


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GRIDS: Cross-modality Surrogate Mapping $\hat{\mathcal{F}}$

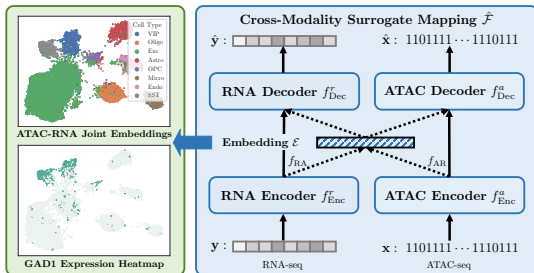
- $\hat{\mathcal{F}}$ modeled by Auto-encoders

$$\mathbf{h}_a^{(i)} = f_{\text{Enc}}^a(\mathbf{W}_{\text{Emb}}^a(\mathbf{x}^{(i)})), \mathbf{h}_r^{(i)} = f_{\text{Enc}}^r(\mathbf{W}_{\text{Emb}}^r(\mathbf{y}^{(i)}))$$

- Learn a joint embedding space to align RNA and ATAC sequences by adversarial training

$$\tilde{\mathbf{h}}_r^{(i)} = f_{\text{AR}}(\mathbf{h}_a^{(i)}), \tilde{\mathbf{h}}_a^{(i)} = f_{\text{RA}}(\mathbf{h}_r^{(i)})$$

- Enables cross-modality generations (e.g. ATAC-seq $\rightarrow$ RNA-seq)



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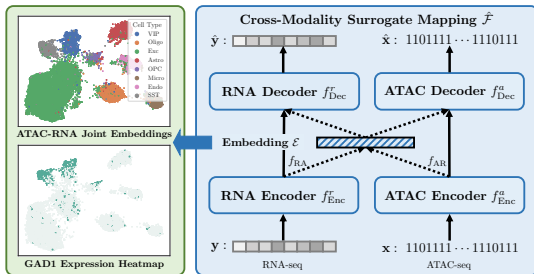
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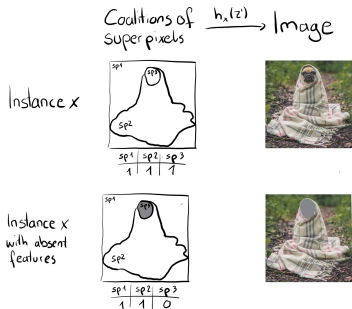
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GRIDS: Global Feature Explanation for Regulatory Redundancy

Use global feature explanation to understand regulatory redundancy

- **Feature Explanation** explains the reason of decision made by the model through adding perturbations on inputs
- Given a target gene in y , how to use global explanation to find a subset r of k indices out of the entire dimension of x ?
- **Local Feature Explanation** seeks to explain individual predictions
 - Instance-wise explanation
- **Global Feature Explanation** seeks to characterize a model's decisions across a population of instances
 - MNIST example (flip model predictions from 8 to 3)



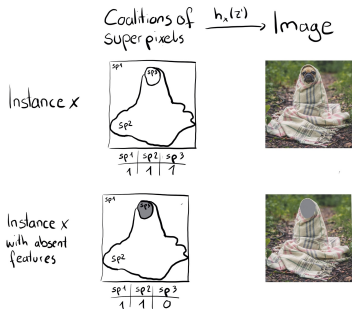
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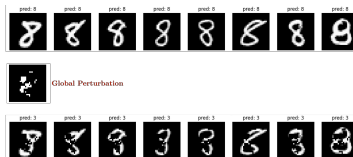
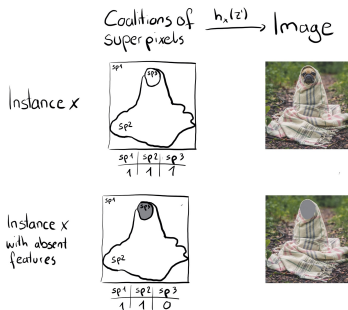
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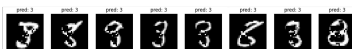
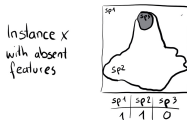
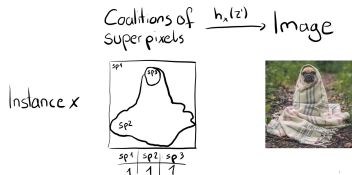
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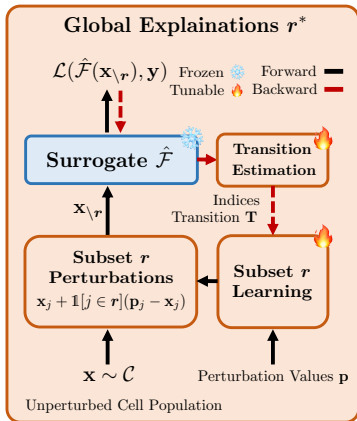
GRIDS: Finding Global Subset Perturbations by End-to-End Learning

Directly applying pervious global feature explanation method is challenging

- ATAC-seq x is **sparse, discrete, and high-dimension**
- Finding a combinatorial subset is too vast to be computationally tractable

Our solution: a global explanation method capable of performing end-to-end training on discrete data

- Optimize over dataset
- Utilize auto-differentiaion
- A unified perturbation operator
 - To make replacement operations be directly optimized as a continuous variable
 - Can be easily extended to different perturbation forms



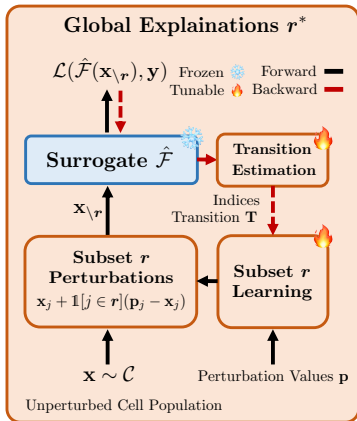
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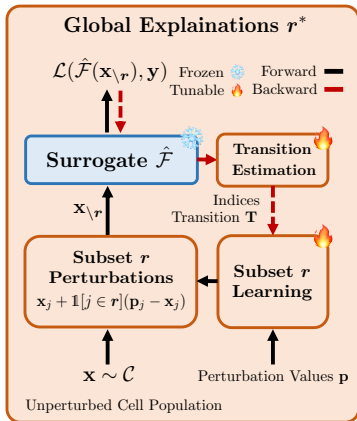
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GRIDS: Understanding Transcription via Global Feature Explanation

Global Explanation Measurement

How much a model's performance degrades over observed samples when features are perturbed

$$r^* = \operatorname{argmin}_r \mathbb{E}_{\mathbf{c} \sim \mathcal{C}} [\mathcal{L}(\hat{\mathcal{F}}(\mathbf{x}_{\setminus r}), \mathbf{y})]$$

r a subset of L peak indices $r = \{r_1, \dots, r_L\}$ to be perturbed

- across a population of cells $\mathcal{C}$

- solution space: $\binom{d_a}{L}$

$\mathbf{x}_{\setminus r}$ denotes the perturbed features indicated by r (i.e, $\mathbf{x}_{\setminus r, r_j} = \mathbf{p}_{r_j}$) (special case, removing features if $\mathbf{p} = \mathbf{0}$)

$\mathcal{L}$ a loss measurement for expected performance degradation

Problem

- The objective involves discrete operation, which is reformulated as $\mathbf{x}_{\setminus r, j} = \mathbf{x}_j + \mathbf{1}[j \in r](\mathbf{p}_j - \mathbf{x}_j)$ to enable auto-differentiation
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GRIDS: Learning Subset Perturbations by Estimating State Transition

Subset Transition Matrix for Gradient Estimation

- gradient w.r.t the input embedding

$$\mathbf{G} = \partial \mathbb{E}_{\mathbf{c} \sim \mathcal{C}} [\mathcal{L}(\hat{\mathcal{F}}(\mathbf{x}_{\setminus r}), \mathbf{y})] / \partial \mathbf{W}_{\text{Emb}}^a(\mathbf{x}_{\setminus r})$$

- construct the transition matrix $\mathbf{T} \in \mathbb{R}^{L \times d_a}$ via first-order approximation
- $\mathbf{T}_{i,j}$ in the matrix represents the advantage value of transitioning from replacing the previous index r_i with the new index j

$$\mathbf{d}_j = \mathbf{G}_j \cdot (\mathbf{W}_{\text{Emb}}^a(\mathbf{p})_j - \mathbf{W}_{\text{Emb}}^a(\mathbf{x})_j)$$
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- Coordinate descent method for updating the subset with one element at a time
- Need to implement a custom PyTorch optimizer

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- $\mathbf{T}_{i,j}$ in the matrix represents the advantage value of transitioning from replacing the previous index r_i with the new index j

$$\mathbf{d}_j = \mathbf{G}_j \cdot (\mathbf{W}_{\text{Emb}}^a(\mathbf{p})_j - \mathbf{W}_{\text{Emb}}^a(\mathbf{x})_j)$$
$$\mathbf{T}_{i,j} = \mathbf{1}[j \notin \mathbf{r}] \mathbf{d}_j - \mathbf{1}[j \neq r_i] \mathbf{d}_{r_i}$$

Perturbation Subset Update Mechanism

- Coordinate descent method for updating the subset with one element at a time
- Need to implement a custom PyTorch optimizer

GRIDS: Learning Subset Perturbations by Estimating State Transition

Subset Transition Matrix for Gradient Estimation

- gradient w.r.t the input embedding

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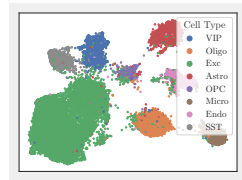
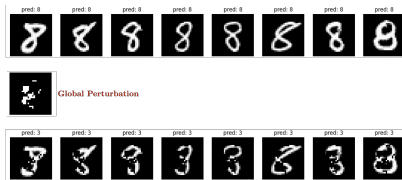
2 Methodology

3 Experiment Results

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Datasets & Experimental Setup

- **MNIST Dataset** To find combinatorial effects found by GRIDS, we conducted experiments on the binary digit classification using MNIST as a start, to better explain our idea.
 - Flip model predictions from 8 to 3
- **Real Single-cell Dataset** We curated a set of deeply-sequenced single-cell multi-modal data from postmortem human pre-frontal cortex (PFC).
 - 10,266 cells with 8 different cell types



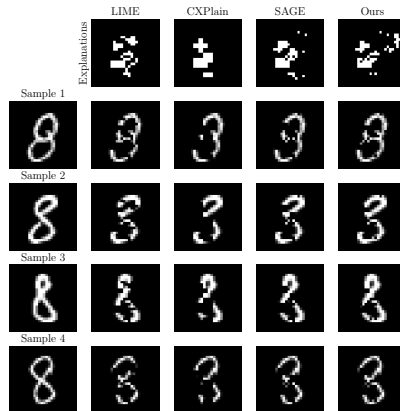
Demo Experiments on MNIST with GRIDS

Experiment Setup and Results

- A pretrained binary classifier (8 vs. 3) that achieves 97.9% accuracy over the test set
- Flip prediction from 8 to 3 by masking $L = 64$ the most important pixels (i.e., $p = 0$)
- GRIDS achieves a combinatorial pattern similar to that of SAGE

Apply perturbations to ATAC-seq x

The pixel here is analogous to the binary entry in the ATAC-seq x , while the binary classifier corresponds to the surrogate $\hat{f}$



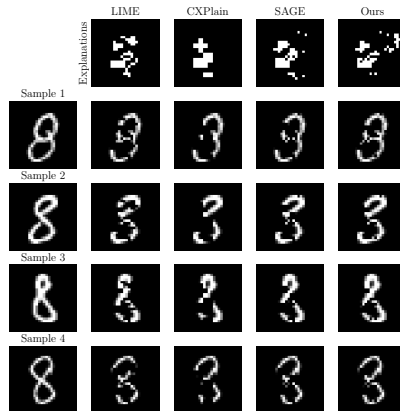
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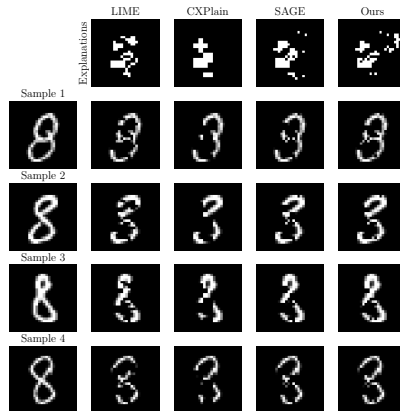
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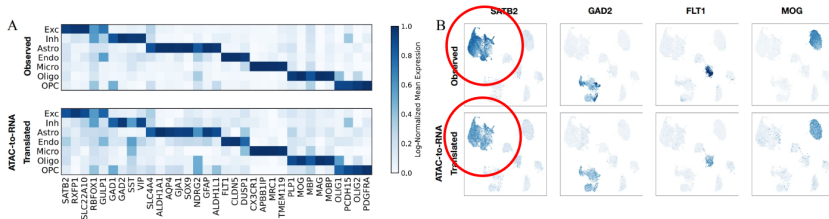
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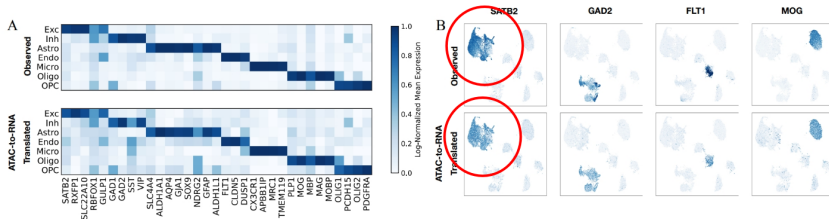
The Surrogate Model $\hat{\mathcal{F}}$ Accurately Models the ATAC-to-RNA Relationship

- Comparing the **mean expressions** between cell types and between the observed and translated cohort
- The averaged expression change (Avg. Δ) and the ratio of expression change against the original value (Rel. Δ)
- The predicted marker gene expression with actual values are pretty same
- UMAP of four key marker genes also similar



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GRIDS consistently outperforms all baselines across each cell type on finding global explanations

- Learning global perturbations for specific marker gene of each cell type
- Among all the model, GRIDS achieves worst avg, which means it can find important CREs

Cell Type	Random		Saliency		SmoothGrad		FIMAP		GRIDS	
	Avg. Δ	Rel. $\Delta(\%)$	Avg. Δ	Rel. $\Delta(\%)$	Avg. Δ	Rel. $\Delta(\%)$	Avg. Δ	Rel. $\Delta(\%)$	Avg. Δ	Rel. $\Delta(\%)$
Astro	-0.085	-0.015	-2.163	-0.601	-2.155	-0.621	-13.502	-4.254	-16.696	-5.837
Endo	-1.073	-0.138	-4.974	-0.372	-9.726	-0.995	-38.997	-9.303	-57.477	-11.816
Micro	-0.012	-0.026	-23.757	-1.545	-32.944	-2.083	-73.752	-6.248	-90.607	-7.671
OPC	+0.823	-0.087	-54.645	-2.338	-48.438	-2.067	-77.167	-6.260	-96.661	-8.256
Oligo	-0.058	+0.026	-0.558	-0.173	-0.939	-0.220	-10.917	-4.252	-16.760	-6.896
SST	+0.159	+0.080	-5.201	-2.006	-5.201	-2.006	-16.453	-5.660	-17.677	-6.365
VIP	+0.012	+0.001	-0.654	-1.189	-0.634	-1.160	-2.732	-3.797	-6.804	-7.195
Avg.	+0.016	-0.021	-12.988	-1.209	-13.519	-1.290	-30.268	-5.367	-39.103	-7.300
Astro	-1.793	-0.533	-15.511	-4.853	-18.505	-6.217	-82.565	-24.766	-100.556	-34.633
Endo	+2.554	+0.468	-46.160	-6.217	-52.383	-7.893	-252.338	-41.790	-259.920	-44.601
Micro	-9.091	-0.490	-131.512	-9.122	-145.561	-10.116	-451.210	-39.695	-470.430	-44.114
OPC	-1.848	-0.165	-193.739	-10.260	-186.235	-9.891	-415.231	-35.687	-392.326	-36.380
Oligo	-1.134	-0.211	-19.809	-6.382	-21.136	-7.630	-69.460	-28.175	-93.518	-38.982
SST	-1.681	-0.615	-33.589	-11.675	-32.275	-11.115	-86.191	-29.198	-93.772	-33.708
VIP	+0.071	+0.002	-4.014	-4.876	-3.872	-4.782	-13.054	-16.757	-19.703	-27.221
Avg.	-1.843	-0.237	-68.620	-7.618	-70.292	-8.212	-202.368	-30.787	-209.583	-36.893

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GRIDS consistently outperforms all baselines across highly-expressed gene sets from two representative cell types

- Top 100 Highly Expressed Genes Expression Changes in VIP and Microglia by masking L important CREs
- GRIDS can also gives global explanation even with a wide range of genes

Type	L	Method	Avg. Δ	Rel. Δ (%)
VIP-100	10	Random	-0.448	-0.009
		Saliency	-18.822	-0.915
		SmoothGrad	-18.424	-0.927
		FIMAP	-56.469	-3.087
		GRIDS	-64.016	-3.827
Microglia-100	10	Random	-0.333	-0.008
		Saliency	-42.372	-1.941
		SmoothGrad	-44.125	-2.073
		FIMAP	-115.092	-5.863
		GRIDS	-141.339	-7.466

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GRIDS can identify CREs with biology insights

- Calculate CRE-to-gene distance with Soft Hit Ratio and Hit Ratio
- Soft Hit Ratio (SHR) measures how many of the reported L CREs are located in this neighborhood
- Hit Ratio (HR) calculates an exact match
- GRIDS's global explanations gives a larger percent of directly interacting CREs verified in experiments

Method	$L = 10$		$L = 128$	
	HR ↑	SHR ↑	HR ↑	SHR ↑
Saliency	0.00	0.00	6.25	18.75
SmoothGrad	0.00	0.00	0.00	18.75
FIMAP	12.50	12.50	18.75	56.25
GRIDS	18.75	25.00	31.25	68.75

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Thanks

Prof. Jing Zhang
Dr. Martin Renqiang Min
Mr. Yaqi Hu

Links

Paper



GitHub



Positions



Zhang Lab at UC Irvine

Thanks for your listening!
Q & A